Create a 30-day learning roadmap.

Context:
- Current Situation: Student
- Current Skills: excel, Canva, powerpoint,
- Goal: to learn the complete Python language
- Available Time: 40min to 1hr
- Experience Level: Beginner
- Preferred Learning Style: Videos & Projects

Include:
- Weekly milestones
- Daily tasks
- Resources
- Projects
- Final outcome

Make it practical and beginner-friendly.

Thought for 73 seconds ›

Here's a practical 30-day roadmap designed for a beginner student who learns best through **videos and hands-on projects**.
You'll go from zero to building a working Python application, all in just **40–60 minutes per day**.

🧭 Overview

- **Goal:** Learn core Python syntax, data structures, OOP basics, file handling, and key libraries (pandas, matplotlib)
- **Style:** Watch → Code → Build
- **Weekly rhythm:** 6 days of learning + 1 review/project day
- **Final outcome:** A solid foundation and a capstone project you can show off

> ⏱ Daily structure: **15–25 min video + 20–35 min practice/coding**

📅 Week 1: Python Foundations & First Programs

Milestone: *Write interactive scripts using variables, strings, numbers, and user input. Build a calculator and a Mad Libs game.*

Day	Topic	Video / Resource	Practice & Mini-Proje
1	Setup & "Hello, World!"	freeCodeCamp Python Course – watch **Intro + Install (0:00–15:00)**	Install Python & VS C triangle shape, and a
2	Variables & Data Types	Same video – **Variables & Data Types (18:00–27:00)**	Create variables for ` type()` to check the
3	Working with Strings	**Strings (27:00–39:00)**	Combine strings, use `.replace()`, f-string
4	Numbers & Math	**Numbers (39:00–48:00)**	Write a script that do input numbers

Day	Topic	Video / Resource	Practice & Mini-Proje
5	Getting User Input	**Getting Input (48:00–57:00)**	Mini-project: **Mad Lil** adjective, noun, etc.,
6	Build a Calculator	**Basic Calculator project (57:00–1:10:00)**	Expand the calculato handle division by ze message
7	🧪 Review & Practice Day	Re-watch any confusing parts	Challenge: Build a **"F** collects info and prin

📅 Week 2: Control Flow & Data Structures

Milestone: *Control program flow with conditionals and loops, and store data in lists, tuples, and dictionaries. Build a quiz game.*

Day	Topic	Video / Resource	Practice & Mini-Proje
8	Lists	freeCodeCamp – **Lists (1:10:00–1:20:00)**	Create a grocery list, items
9	List Functions & 2D Lists	**List Functions (1:20:00–1:30:00)**	Make a 2D grid (tic-ta `len()`, `sort()`, `pop`
10	Tuples	**Tuples (1:30:00–1:36:00)**	Create a tuple of coo (and see the error)
11	Dictionaries	**Dictionaries (1:36:00–1:48:00)**	Build a dictionary of s through keys/values
12	If Statements & Comparisons	**If Statements (1:48:00–2:02:00)**	Write a simple login s username/password
13	While Loops	**While Loop (2:02:00–2:10:00)**	Mini-project: **Numbe** random secret numb
14	For Loops & Week Project	**For Loop (2:10:00–2:20:00)**	**Weekly project: Quiz** questions & answers,

📅 Week 3: Functions, Files & Error Handling

Milestone: *Write your own functions, read/write files, handle errors, and use built-in modules. Build a persistent To-Do list.*

Day	Topic	Video / Resource	Practice & Mini-Proj
15	Functions Basics	Corey Schafer - Functions (watch first 20 min)	Write a function to g square a number
16	Parameters & Return Values	Finish the Functions video	Build a **temperature** functions with retur
17	Try / Except (Error Handling)	Corey Schafer - Error Handling	Add error handling t Week 1 (non-numbe
18	Reading Files	Corey Schafer - File Objects (read part)	Read a text file (poe or lines

Day	Topic	Video / Resource	Practice & Mini-Pro
19	Writing & Appending Files	Same video – writing section	**Mini-app: CLI Note** with a timestamp
20	Modules & `pip`	Corey Schafer - Modules (first half)	Explore `math` , `ran` pip. Build a **random**
21	🧪 Combine & Project	Review any needed videos	**Weekly project: To-** add/remove/list tasl it persists between

📅 Week 4: OOP, Data Analysis & Final Project

Milestone: *Understand classes and objects, use pandas for data analysis, create plots with matplotlib, and start your capstone.*

Day	Topic	Video / Resource	Practice & Mini-Pr
22	Classes & Objects	Corey Schafer - OOP Part 1	Create a `Dog` clas a `bark()` method
23	`__init__` & Instance Methods	Continue the video	Build a simple **Ban** withdraw, check b
24	Inheritance Basics	Corey Schafer - Inheritance (first 20 min)	Create a `SavingsA` BankAccount, add
25	Intro to pandas	Corey Schafer - Pandas Part 1 (Load & inspect data)	Load a CSV (e.g., DataFrame, use `.I` `.info()`
26	Data Analysis with pandas	Same series (filtering & grouping)	Filter rows, group I averages (great fo
27	Intro to matplotlib	Corey Schafer - Matplotlib Part 1 (up to bar charts)	Plot a line chart an DataFrame
28	🧩 Connect pandas + matplotlib	Practice with your own dataset	**Weekly milestone** monthly expenses and a bar chart of

🏁 Days 29–30: Capstone Project & Wrap-up

Choose **one** project (or mix them), build it from scratch, and apply everything you've learned.

💰 Project Option A: Personal Finance Tracker

- Let the user enter expenses (amount, category, date)
- Store entries in a CSV file
- When the user runs a "report", read the CSV with pandas
- Show total per category and a bar chart

🎓 Project Option B: Student Grade Manager

- Enter student names and grades for multiple subjects

- Save to a CSV
- Load and calculate each student's average
- Plot a bar chart of class averages per subject

Day 29: Plan, set up the file I/O, implement the core logic
Day 30: Add analysis (pandas) and visualization (matplotlib), test, and maybe add a small README

📚 Key Resources (All Free)

Resource	Link
Main beginner course (Weeks 1–2)	freeCodeCamp – Learn Python (4.5h)
Functions, OOP, Files, pandas, matplotlib	Corey Schafer's Python Playlist
Quick syntax practice	W3Schools Python
Online coding environment (no install)	Replit

🎯 Final Outcome

After 30 days you'll be able to:

- Write clean Python scripts with functions, loops, and error handling
- Read, process, and write files
- Use Python's data structures confidently
- Understand the basics of object-oriented programming
- Analyze and visualize data with pandas & matplotlib (a natural upgrade from Excel)
- Complete a **capstone project** that ties everything together and serves as a portfolio piece

"Learn the complete Python language" is a journey that takes much longer than 30 days, but this roadmap gives you the **practical, working core** – and the ability to keep learning anything else you need.

Let me know if you'd like a Google Sheets version or daily checklists to track your progress!

This response is AI-generated, for reference only.

Create a PDF file of this response

I can't directly generate or send you a PDF file, but here are the **quickest ways** to turn the roadmap into a PDF yourself:

🔷 Option 1: Browser Print (Easiest – 30 seconds)

1. **Select my entire roadmap response** (from "Here's a practical 30-day..." to the end).
2. Right-click and choose **Print** (or press `Ctrl+P` / `Cmd+P`).
3. In the print dialog, choose **"Save as PDF"** as the destination.
4. (Optional) Under "More settings," check **"Background graphics"** for better formatting.
5. Click **Save** — you'll have a clean PDF.